

Fasudil inhibits lysophosphatidic acid-induced invasiveness of human ovarian cancer cells.

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Ovarian cancer is known to be highly invasive. The poor prognosis of advanced ovarian cancer comes from increased invasiveness of human ovarian cancer cells. The lysophosphatidic acid (LPA)/Rho/Rho-associated kinase (ROCK) pathway is intimately involved in the course of ovarian cancer progression, and the inhibition of this pathway attenuates ovarian cancer invasiveness. Fasudil (1-[5-isoquinolinesulfonyl]-homopiperazine; HA-1077) is a drug that has been in clinical use in Japan for the prevention of vasospasm after subarachnoid hemorrhage and is known to be a potent ROCK-specific inhibitor. In this study, we examined the effect of fasudil on LPA-induced invasiveness of human ovarian cancer cells to explore the potential of fasudil as an anticancer agent against ovarian cancer. Fasudil induced changes in cell morphology but not in cell viability. Fasudil significantly inhibited LPA-induced invasion and motility of human ovarian cancer cells in a dose-dependent manner. Furthermore, fasudil caused the loss of intracellular cytoskeletal rearrangement, which is necessary for cell motility, such as stress fiber formation and focal adhesion assembly. Fasudil suppressed LPA-induced tyrosine phosphorylation of paxillin, a representative focal adhesion protein, and serine phosphorylation of myosin light chain, which are essential for the process for cell migration. These findings showed that fasudil attenuated the invasiveness of human ovarian cancer cells via inhibition of the LPA/Rho/ROCK pathway. In SKOV-3ip1 ovarian cancer xenografts, intraperitoneal treatment with fasudil significantly reduced tumor burden and ascites formation. Our findings suggest that fasudil might be useful to prevent the progression of ovarian cancer in clinical settings.

Fasudil and its analogs: a new powerful weapon in the long war against central nervous system disorders?

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Rho kinase (ROCK) plays a critical role in actin cytoskeleton organization and is involved in diverse fundamental cellular functions such as contraction and gene expression. Fasudil, a ROCK inhibitor, has been clinically applied since 1995 for the treatment of subarachnoid hemorrhage (SAH) in Japan. Increasing evidences indicate that fasudil could exhibit markedly therapeutic effect on central nervous system (CNS) disorders, such as Alzheimer's disease. This article summarizes results from supporting evidence for the potential therapy for fasudil against a variety of CNS diseases. And the properties of its analogs are also summarized. Current therapies against CNS disorders are only able to attenuate the symptoms and fail in delaying or preventing disease progression and new approaches with disease-modifying activity are desperately needed. The dramatic effects of fasudil in animal models and/or clinical applications of CNS disorders make it a promising strategy to overcome CNS disorders in human beings. Given the complex pathology of CNS disorders, further efforts are necessary to develop multifunctional fasudil derivatives or combination strategies with other drugs in order to exert more powerful effects with minimized adverse effects in the combat of CNS disorders.

Fasudil, a Rho-Kinase Inhibitor, Exerts Cardioprotective Function in Animal Models of Myocardial Ischemia/Reperfusion Injury: A Meta-Analysis and Review of Preclinical

Evidence and Possible Mechanisms.

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Fasudil, a Rho-kinase inhibitor, has shown outstanding therapeutic effects against cerebral vasospasm after subarachnoid hemorrhage (SAH) in humans. Studies show various biological effects of fasudil in the cardiovascular system. We conducted a preclinical systematic review to determine the efficacy and possible mechanisms of fasudil on animal models of myocardial ischemia/reperfusion (I/R) injury. Nineteen studies involving 400 animals were identified after searching 8 databases for articles published till June 2018. The methodological quality was assessed by the Collaborative Approach to Meta-Analysis and Review of Animal Data from Experimental Studies (CAMARADES) 10-item checklist. The data were analyzed using Rev-Man 5.3 software, and the score of study quality ranged from 3 to 6 points. Compared to the control group, fasudil treated animals showed reduced myocardial infarct size ($P < 0.05$), lower levels of cardiac enzymes ($P < 0.05$) and cardiac troponin T ($P < 0.05$), improved systolic and diastolic functions ($P < 0.05$), and increased degree of decline in the ST-segment ($P < 0.05$). The possible mechanisms of fasudil action against myocardial I/R injury are improvement in coronary vasodilation, inhibition of apoptosis and oxidative stress, relieving inflammation, and reduction in endoplasmic reticulum stress and metabolism. In conclusion, fasudil exerts a cardio-protective function through multiple signaling pathways in animal models of myocardial I/R injury.

Therapeutic potential of rho-kinase inhibitors in cardiovascular diseases.

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Rho-kinase is a signaling molecule that occurs downstream of the small GTPase Rho, which mediates various cellular functions. The Rho/Rho-kinase pathway plays an important role in pathophysiology and progression of various cardiovascular diseases such as hypertension, coronary vasospasm, angina pectoris, and restenosis after percutaneous coronary intervention, all of which are related to arteriosclerosis/atherosclerosis changes of the vasculature. Activation of the Rho/Rho-kinase pathway contributes to inflammatory and proliferative changes of the blood vessels and affects cardiac myocytes. Evidence from in vitro and in vivo studies suggests that Rho-kinase inhibitors have beneficial effects on cardiovascular diseases, particularly arteriosclerosis and coronary vasospasm. Furthermore, activation of the Rho/Rho-kinase pathway contributes to blood pressure regulation via the central sympathetic nervous system. There is evidence to suggest that Rho-kinase is involved in angiotensin II-induced cardiac hypertrophy and endothelial dysfunction, and preliminary data indicate that inhibition of Rho-kinase may be beneficial in vascular disorders such as pulmonary arterial hypertension and erectile dysfunction. Fasudil is currently the only Rho-kinase inhibitor available for clinical use and it is approved in Japan for the prevention of vasospasm in patients with subarachnoid hemorrhage. Emerging clinical data have shown that oral fasudil 80 mg three times daily is effective in preventing myocardial ischemia in patients with stable angina pectoris. Rho-kinase represents a new target for the management of cardiovascular diseases and further studies are needed to define the therapeutic potential of Rho-kinase inhibitors.

Multichannel near-infrared spectroscopy as a tool for assisting intra-arterial fasudil therapy for diffuse vasospasm after subarachnoid hemorrhage.

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Diffuse cerebral vasospasm following aneurysmal subarachnoid hemorrhage (SAH) refractory to medical management can be treated with intra-arterial administration of vasodilators, but valid bedside monitoring for the diagnosis and therapeutic assessment is poorly available. We demonstrate the successful application of regional cerebral oxygen saturation (rSO(2)) monitoring with multichannel near-infrared spectroscopy (NIRS) in assisting intra-arterial infusions of fasudil hydrochloride to a patient suffering from post-SAH vasospasm in the distal vascular territories. A 63-year-old man presented with SAH and intracerebral hematoma due to ruptured right middle cerebral artery aneurysm developed aphasia and right-sided weakness on day 9 after SAH onset. Delayed cerebral ischemia attributable to diffuse vasospasm in the distal territories of the left anterior and middle cerebral arteries was suspected. Since the symptoms persisted despite maximal hyperdynamic therapy with dobutamine, intra-arterial fasudil treatment in the setting of rSO(2) monitoring including the spasm-affected vascular territory with four-channel flexible NIRS sensors was subsequently performed. Decreased and fluctuating rSO(2) in angiographically documented vasospastic territories increased immediately after intra-arterial fasudil infusion in accordance with relief of vasospasm that correlated with neurological improvement. The procedure was repeated on day 11 since the effect was transient and neurological deterioration and reduction of rSO(2) recurred. The deficits resolved accompanied by uptake and maintenance of rSO(2) following the intra-arterial fasudil, resulting in favorable functional outcome. Continuous rSO(2) monitoring with multichannel NIRS is a feasible strategy to assist intraarterial fasudil therapy for detecting and treating the focal ischemic area exposed to diffuse vasospasm.

An overview of pharmacotherapy for cerebral vasospasm and delayed cerebral ischemia after subarachnoid hemorrhage.

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Introduction: Survival from aneurysmal subarachnoid hemorrhage has increased in the past few decades. However, functional outcome after subarachnoid hemorrhage is still suboptimal. Delayed cerebral ischemia (DCI) is one of the major causes of morbidity. **Areas covered:** Mechanisms underlying vasospasm and DCI after aneurysmal subarachnoid hemorrhage and pharmacological treatment are summarized in this review. **Expert opinion:** Oral nimodine, an L-type dihydropyridine calcium channel blocker, is the only FDA-approved drug for the prevention and treatment of neurological deficits after aneurysmal subarachnoid hemorrhage. Fasudil, a potent Rho-kinase inhibitor, has also been shown to improve the clinical outcome and has been approved in some countries for use in patients with aneurysmal subarachnoid hemorrhage. Although other drugs, including nicardipine, cilostazol, statins, clazosentan, magnesium and heparin, have been expected to have beneficial effects on DCI, there has been no convincing evidence supporting the routine use of those drugs in patients with aneurysmal subarachnoid hemorrhage in clinical practice. Further elucidation of the mechanisms underlying DCI and the development of effective therapeutic strategies for DCI, including combination therapy, are necessary to further improve the functional outcome and mortality after subarachnoid hemorrhage.

Prenatal and Postnatal Ocular Abnormalities Following Congenital Zika Virus Infections: A Systematic Review.

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To assess fetal and neonatal eyes abnormalities and their progression during the last ZIKV outbreak and summarize learned lessons. A systematic review and meta-analysis was conducted by a team of obstetricians and ophthalmologists. Studies reporting ocular abnormalities during the prenatal ($n = 5$) and postnatal ($n = 24$) periods were included in the analysis. In the prenatal period, the most common ocular findings were intraocular calcification cases (4/6, 66.6%) and microphthalmia (3/6, 50%). Postnatal ocular abnormalities of congenital ZIKV infection were described after birth in 479 cases. Among them microphthalmia was reported in 13 cases (13/479, 2.7%). Posterior segment (retina and optic nerve) was the most affected structure, consisting of pigmentary changes (229/479, 47.8%), macular chorioretinal atrophy (216/479, 45%), optic nerve atrophy (181/479, 37.8%), increased cup-to-disk ratio (190/479, 39.6%), optic nerve hypoplasia (93/479, 19.4%), vascular changes (26/479, 5.4%), and retinal coloboma (20/479, 4.1%). The anterior segment was involved in 4.6% (22/479) of cases, including cataract (9/479, 1.8%), lens subluxation (1/479, 0.2%), iris coloboma (5/479, 1%), and congenital glaucoma (7/479, 1.4%). These ocular anomalies were isolated in one case (1/479, 0.2%) and multiple anomalies were found in the other cases. Long-term visual disorders have been described, with no possible improvement and even a worsening of some of the ocular anomalies previously observed. No reactivation of ocular lesions was observed. This review highlights the severe ocular abnormalities associated with congenital ZIKV infections. The importance of multidisciplinary communication between the obstetrician, the maternal-fetal medicine specialist, and the ophthalmologist is emphasized. This systematic review was registered with the International Prospective Register of Systematic Reviews (PROSPERO), registration440 188.

Epidemiological analysis of congenital glaucoma: a national scenario.

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Congenital glaucoma is a disease that involves increased intraocular pressure and can result in irreversible visual deterioration. The study of epidemiology allows the delineation of the characteristics associated with patients and specific risk factors. The objective of this study was to examine epidemiological trends, place of residence, duration of gestation, sex, and race of the newborn diagnosed with congenital glaucoma in Brazil. Data from SINASC (National Live Birth System) were used to analyze the period from 2017 to 2021 in Brazil. Linear regression and analysis of variance tests were employed to assess significance. The statistical significance was determined by $p < 0.05$. A total of 47 cases of congenital glaucoma were identified in Brazil during the study period, with the highest incidence between the years of 2018 and 2021. The analysis of the distribution indicated that the states with the highest incidence were São Paulo, followed by Rio Grande do Sul and Pernambuco. Approximately 60% of cases occurred in male individuals, compared with 19 female cases. The ethnic analysis showed the highest incidence among whites and mixed. Regarding the length of pregnancy, statistical differences were observed between newborns of different periods of gestation. Infants born from pregnancies lasting between 28 and 31 weeks and 32 and 36 weeks were significant when analyzed with the group between 37 and 41 weeks. Studies on the mechanisms of congenital glaucoma seek to improve knowledge about the disease. Epidemiological evaluation is essential for identifying demographic and clinical patterns of the disease.

Trends in Childhood Glaucoma Prevalence and Incidence in South Korea, 2002-2019: A Nationwide Population-Based Study.

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This nationwide analysis identified the prevalence and incidence of childhood glaucoma for an 18-year period. The prevalence and incidence of primary congenital glaucoma showed increasing trends. Juvenile open angle glaucoma, meanwhile, showed a decreasing tendency. We aimed to determine the trends in the prevalence and incidence of childhood glaucoma in the entire population of South Korea. A nationwide retrospective cohort study was performed with an age-specific and sex-specific population of South Korea. The Korean National Health Insurance Service claims database for 2002 to 2019 was accessed to identify cases of ophthalmologist-confirmed primary childhood glaucoma [ie, primary congenital glaucoma (PCG) and juvenile open angle glaucoma (JOAG)]. Incidence for PCG was estimated for a same-birth-year population, while that for JOAG was estimated using age-specific and sex-specific population figures. To verify the glaucoma cases, we also analyzed the diagnostic codes as well as any information on medication prescriptions and/or ocular-surgery history. During the 18-year observational period, totals of 505 and 7538 patients were diagnosed with PCG and JOAG, respectively. The mean prevalences of PCG and JOAG were 3.96 ± 0.72 and 14.17 ± 5.18 , respectively. The prevalence of PCG showed an overall increasing trend during the study period, but the pattern was not significant ($\beta = 0.049$, $P = 0.143$); that of JOAG, meanwhile, showed a significant decreasing tendency ($\beta = -0.713$, $P = 0.001$). PCG prevalence showed no difference between urban and rural areas, but JOAG showed a higher prevalence in rural areas ($P < 0.001$). As for mean incidence, the rates for PCG and JOAG were 1.54 ± 0.49 and 5.02 ± 1.95 (per 100,000 person-years), respectively, and were higher in males ($P < 0.001$ and $P = 0.013$). This study identified childhood glaucoma prevalence and incidence in a general population of East Asian ethnicity. This data could help to promote a better understanding of the typical epidemiological features and clinical courses of childhood glaucoma patients.

CYP1B1 and MYOC variants in neonatal-onset versus infantile-onset primary congenital glaucoma.

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To compare CYP1B1 and MYOC variants in a cohort of neonatal-onset (NO) and infantile-onset (IO) primary congenital glaucoma (PCG). This prospective observational study included 43 infants with PCG (14 NO and 29 IO) presenting between January 2017 and January 2019 with a minimum 1-year follow-up. CYP1B1 and MYOC genes were screened using Sanger sequencing with in-silico analysis of the variants using Polymorphism Phenotyping v.2 and Protein Variation Effect Analyser platforms. Allelic frequency was estimated using Genome Aggregation Database (gnomAd). Disease presentation and outcome were correlated to the genetic variants in both groups. Babies with CYP1B1 mutations had more severe disease at presentation and worse outcomes. Six of 14 (42.8%) NO glaucoma and 5 of 29 (17.2%) IO harboured CYP1B1 mutations. Five of six babies in the NO group and three of five in the IO group harboured the variant c.1169G>A, [p.R390H]. They required more surgeries and had a poorer outcome. On in-silico analysis c.1169G>A, [p.R390H] scored very likely pathogenic. Two patients in the IO group who had the c.1294C>G, [p.L432V] variant had a good outcome. Five of 14 NO-PCG and 8 of 29 IO-PCG harboured the variant c.227G>A, [p.R76K] in the MYOC gene, which was scored benign by in-silico analysis, and was also found in 2 of 15 normal controls. Patients with CYP1B1 pathogenic variants had a poorer outcome than those without. We found more NO PCG babies with CYP1B1 mutations compared

with IO PCG. This may be one of the reasons for NO PCG having a poorer prognosis compared with IO PCG.
